Sham Satish Thakare

Candidate for PhD in Quantum Science & Engineering (QSE) · Quantum Information, Circuit Compilation & Complexity

Grade O in Quantum AI (SPPU Course 417523A) · Gold Medalist in Mathematics (2024) · Pre-Doc, Capital One Systems Research

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QUANTUM SCIENCE RESEARCH INTERESTS

Quantum Information Theory: Multi-qubit density matrix formalism (rho in S(H), rho >= 0, Tr(rho)=1), von Neumann entropy S(rho) = -Tr(rho log rho), quantum fidelity bounds, and trace-preserving completely positive (CPTP) noise maps.
Quantum Circuit Compilation & Optimization: Stabilizer code projections (Gottesman-Knill theorem), surface code syndrome extraction cycles, tensor network contractions (MPS/PEPS), and error mitigation via zero-noise extrapolation (ZNE).
Quantum Complexity Theory: Complexity separations (BPP $\subseteq$ BQP $\subseteq$ PSPACE), quantum query complexity speedups, and non-local games.

EDUCATION & ACADEMIC DISTINCTIONS

APCOER Pune (Savitribai Phule Pune University), Pune, India — 2020 – 2024

Bachelor of Engineering (B.E.) in AI & DS | CGPA: 8.70 / 10.00 (First Class with Distinction) — Final Year SGPA: 9.53 / 10.00 (Rank 1)

- **Official University Transcript Distinction: Course 417523A: Quantum Artificial Intelligence — Grade O (Outstanding, 10.0/10.0, 30 GP)** on official SPPU transcript (Ref: EXAM./TRANS./BEN241203044).
- **Gold Medal in Mathematics (2024):** Conferred by university leadership for highest academic standing across mathematical and quantitative curricula.
- **Grade O Technical Coursework:** Quantum AI, Distributed Computing, Artificial Neural Networks, Operating Systems, Cyber Security, Capstone Senior Thesis, Industry Internship.
- **Peer Mentorship:** Appointed Peer Academic Mentor in Discrete Mathematics and Quantum Information Foundations at APCOER Pune.

QUANTUM COMPUTATIONAL & THEORETICAL CONTRIBUTIONS

- **Stabilizer Circuit Formalisms & Gottesman-Knill Analysis:** Led student workshops analyzing stabilizer code projections (S_i |psi> = |psi>) and symplectic representation of Pauli operations, demonstrating classical polynomial-time simulation bounds of Clifford group operations.
- **Tensor Network Contractions & Noise Modeling:** Modeled shallow quantum circuit contractions and simulated depolarizing and amplitude-damping noise channels using Qiskit and Cirq, assessing fidelity decay under zero-noise extrapolation (ZNE).
- **Mathematical State Overlap & Entropy Derivations:** Derived analytical bounds for conditional predictive entropy and state distinguishability under CPTP operations, establishing connections between classical stochastic processes and quantum state measurement.

RESEARCH & PROFESSIONAL EXPERIENCE

Capital One Systems Research, Pune, India — 06/2024 – Present

Pre-Doctoral Researcher | Mentorship: Yuri Shelke (Distinguished Engineer)

- Investigated high-performance C++20 state machines and low-latency data structures applicable to quantum circuit scheduling harnesses.
- Formulated mathematical convergence bounds on stochastic policy updates, presenting analytical proofs on credit assignment geometry.

APCOER Pune (Savitribai Phule Pune University), Pune, India — 06/2023 – 05/2024

Peer Academic Mentor in Quantum Information Foundations

Conducted tutorial sessions for undergraduates on Hilbert space linear algebra, spectral decomposition, density operators, and quantum logic gates.

TARGET HARVARD QSE FACULTY ALIGNMENTS

Prof. Mikhail Lukin (Physics / QSE): Quantum simulation, neutral-atom quantum processors, and logical qubit error correction.

Prof. Boaz Barak (SEAS / CS / QSE): Quantum complexity theory, BQP vs BPP bounds, and theoretical separations.

Prof. Susanne Yelin (Physics / QSE): Cooperative quantum dynamics, quantum memories, and quantum information processing.